1. Import Libraries

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification

from google.colab import files
uploaded = files.upload()
```

Choose Files train.csv

train.csv(text/csv) - 61194 bytes, last modified: 4/18/2026 - 100% done

```
Saving train.csv to train.csv
```

2. Load Dataset

```
df = pd.read_csv('train.csv')

df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Tick
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 175
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/(31012
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	1138
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	3734

Next steps: Generate code with df New interactive sheet

3. Data Cleaning

```
# Check missing values
df.isnull().sum()

# Fill Age with mean
df['Age'] = df['Age'].fillna(df['Age'].mean())

# Fill Embarked with mode
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].m

# Verify
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	0

dtype: int64

4. Feature Encoding

```
# Convert Gender to numeric
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})

# Convert Embarked to numeric
df['Embarked'] = df['Embarked'].map({'S': 0, 'C': 1, 'Q': 2})
```

5. Feature Selection

```
features = ['Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare']

X = df[features]
y = df['Survived']
```

6. Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

7. Logistic Regression Model

```
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

8. Model Evaluation

```
print("Accuracy:", accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.8100558659217877

Confusion Matrix:
[[92 13]
 [21 53]]

Classification Report:
              precision    recall  f1-score   support

           0       0.81      0.88      0.84       105
           1       0.80      0.72      0.76        74

    accuracy                           0.81       179
   macro avg       0.81      0.80      0.80       179
weighted avg       0.81      0.81      0.81       179
```

9. Conclusion

- Logistic Regression can effectively classify survival outcomes
- Features like gender and passenger class strongly influence survival
- Model achieved good accuracy indicating reliable predictions